

Executive Customer Intelligence Report

Automated Generation of E-Commerce Transactional & Behavioral EDA

1. Executive Summary

ShopSense Analytics has processed three years of historical customer transactional and clickstream event data. This automated diagnostic report highlights our primary customer traits, overall revenue growth patterns, and key retention metrics. These inputs serve as the structural baseline for our churn prediction, forecasting, and clustering pipelines.

Total Customer Cohort	1,000
Historical Transaction Count	19,307
Average Customer Age	35.0 years (Median: 35.0)
Observed Customer Churn Rate	22.90% (229 churned)
Total Return-Adjusted Revenue	INR 107,997,228.55

2. Churn & Demographic Insights

Demographic distributions show clear trends when stratified by our customer churn labels. Below are the churn rates segmented by key categorical acquisition features and premium statuses:

Premium Member Churn Rate	23.58%
Non-Premium Member Churn Rate	22.72%

Churn Rate by Acquisition Channel:

Email	19.54%
Organic	22.58%
Paid_search	24.80%
Referral	24.34%
Social_media	21.46%

3. Monthly Revenue Trend

The graph below depicts the return-adjusted monthly revenue trend across the observed historical timeframe. Significant seasonal multipliers are visible in the fourth quarter (October through December), driving a 1.5x-2x increase in monthly sales volumes.

